

Scenario-Based Questions

1. Low Contrast Image

A grayscale image appears dull with very little difference between object and background intensities.

(a) Interpret the scenario and explain the cause of low contrast

A low contrast image is a grayscale image where the object and the background have very similar intensity values. As a result, the image looks dull, faded, and unclear, making it difficult to distinguish different objects.

Causes of low contrast:

- Poor or insufficient lighting during image capture.
- Limited camera sensor capability.
- Haze, fog, or shadows.
- Improper exposure settings.
- Pixel intensity values are concentrated within a small range.

(b) Identify the key issues affecting image visibility

The main issues are:

- Poor distinction between object and background.
- Blurred or unclear edges.
- Important details are difficult to identify.
- Reduced image quality.
- Difficult for both human observation and computer vision algorithms to detect features accurately.

(c) Apply an appropriate enhancement technique

Technique: Histogram Equalization

Steps:

1. Read the grayscale image.
2. Calculate the image histogram.
3. Redistribute the intensity values over the full range (0–255).
4. Generate the enhanced image.

Result:

- Dark regions become clearer.
- Bright regions become more visible.
- Overall image contrast increases.
- Objects are easier to distinguish from the background.

(d) Justify your choice

Histogram Equalization is the most suitable technique because:

- It spreads pixel intensity values across the entire grayscale range.
- It improves the visibility of hidden details.
- It enhances the contrast automatically without changing the image content.
- It is simple, fast, and widely used in medical imaging, satellite images, and computer vision applications.

Therefore, it is the best choice for enhancing a low contrast grayscale image.

(e) Clear and structured explanation

Problem:

The grayscale image has very low contrast, making the object and background appear almost the same.

Issues:

- Poor visibility
- Weak edges

- Hidden details
- Difficult object identification

Solution:

Apply Histogram Equalization to redistribute pixel intensity values and improve contrast.

Outcome:

- Better brightness distribution
- Clear object boundaries
- Improved image quality
- Easier feature detection and analysis

2. Salt-and-Pepper Noise

A scanned document contains salt-and-pepper noise affecting readability.

(a) Explain the scenario and characteristics of this noise

A **salt-and-pepper noise** image contains **random black (pepper) and white (salt) pixels** caused by errors during image acquisition, transmission, or scanning. In a scanned document, this noise reduces readability by adding unwanted dots.

Characteristics:

- Appears as random black and white pixels.
- Affects only some pixels.
- Makes text and image details difficult to read.
- Common in scanned documents and digital images.

(b) Identify the key issues impacting image quality

The main issues are:

- Reduced readability of the document.
- Loss of important image details.
- Presence of unwanted black and white dots.
- Difficulty in text recognition (OCR).
- Poor visual quality.

(c) Apply an appropriate filtering technique

Technique: Median Filter

Steps:

1. Select a small neighborhood (e.g., 3×3 window).
2. Arrange the pixel values in ascending order.
3. Replace the center pixel with the **median** value.
4. Repeat for the entire image.

Result:

- Removes salt-and-pepper noise.
- Preserves image edges.
- Produces a smoother and clearer image.

(d) Justify why this method is more suitable than alternatives

The **Median Filter** is the best choice because:

- It effectively removes salt-and-pepper noise.
- It preserves edges better than the Mean (Average) Filter.
- It does not blur important details.
- It improves readability while maintaining image quality.

Therefore, the Median Filter is more suitable than average or Gaussian filters for this type of noise.

(e) Clear and structured explanation

Problem:

The scanned document contains random black and white pixels (salt-and-pepper noise), making it difficult to read.

Issues:

- Poor readability
- Random black and white dots
- Loss of image quality
- Difficult text recognition

Solution:

Apply a **Median Filter** to remove the noise while preserving edges.

Outcome:

- Noise-free image
- Clear and readable document
- Preserved text edges
- Improved image quality

3. Blurred Edges

An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpret the scenario and explain why edges are blurred

The image has been processed to remove noise, but the filtering has made it **overly smooth**. As a result, **edges and fine details become blurred** because the filter averages neighboring pixel values.

Why edges are blurred:

- Strong smoothing filters remove both **noise and important edge information**.
- High-frequency components (edges) are reduced along with the noise.
- Excessive filtering causes loss of sharpness.

(b) Identify key issues related to smoothing and edge loss

The main issues are:

- Blurred object boundaries.
- Loss of fine details.
- Reduced image sharpness.
- Difficulty in object detection and segmentation.
- Lower overall image quality.

(c) Apply an appropriate technique to restore edges

Technique: Sharpening using Unsharp Masking (High-Boost Filtering)

Steps:

1. Blur the original image slightly.
2. Subtract the blurred image from the original to obtain edge details.
3. Add the edge details back to the original image.
4. Obtain a sharper image with restored edges.

Result:

- Sharper edges.
- Better object boundaries.
- Improved image clarity without significantly increasing noise.

(d) Justify your decision

Unsharp Masking is the best choice because:

- It enhances edges while preserving most image details.
- It restores sharpness lost during smoothing.
- It improves visual quality without greatly amplifying noise.
- It is widely used in image enhancement and computer vision.

Therefore, it is more suitable than applying additional smoothing, which would blur the image further.

(e) Clear and structured explanation

Problem:

Noise removal has made the image too smooth, causing blurred edges.

Issues:

- Blurred edges
- Loss of fine details
- Reduced sharpness
- Poor object detection

Solution:

Apply **Unsharp Masking (High-Boost Filtering)** to enhance and restore edges.

Outcome:

- Sharper edges
- Clear object boundaries
- Improved image quality
- Better feature detection

4. Simple Segmentation

An image contains objects clearly distinguishable from the background based on intensity.

(a) Interpret the scenario and explain segmentation requirements

The image contains **objects that are clearly different from the background based on pixel intensity**. This means the object pixels have noticeably higher or lower intensity values than the background, making them easy to separate.

Segmentation requirement:

- Divide the image into **object** and **background**.
- Separate the region of interest accurately.
- Produce a binary image for further analysis.

(b) Identify key features enabling separation

The main features are:

- Clear difference in pixel intensity.
- High contrast between object and background.
- Well-defined object boundaries.
- Minimal noise in the image.

(c) Apply a suitable segmentation technique

Technique: Thresholding (Global Thresholding)

Steps:

1. Select a threshold value (**T**).
2. Compare each pixel intensity with **T**.
3. If the pixel value $\geq T$, classify it as the **object**.
4. If the pixel value $< T$, classify it as the **background**.
5. Generate a binary image.

Result:

- Objects are separated from the background.
- A clear binary segmented image is obtained.

(d) Justify your decision

Global Thresholding is the best choice because:

- The object and background have a clear intensity difference.
- It is simple, fast, and easy to implement.
- It provides accurate segmentation when lighting is uniform.

- It requires low computational cost.

Therefore, Global Thresholding is more suitable than complex segmentation methods for this scenario.

(e) Clear presentation

Problem:

The objects are clearly distinguishable from the background using intensity values.

Issues:

- Need accurate object-background separation.
- Preserve object boundaries.
- Minimize misclassification.

Solution:

Apply **Global Thresholding** to convert the grayscale image into a binary image.

Outcome:

- Clear object extraction.
- Accurate segmentation.
- Improved image analysis.
- Easy feature detection.

5. Combined Requirement

An industrial inspection system requires noise removal while preserving important edges for defect detection.

(a) Interpret the scenario and define the combined requirement

An **industrial inspection system** must detect defects accurately. The image contains noise, but important edges must be preserved because defects are usually identified from object boundaries.

Combined requirement:

- Remove image noise.
- Preserve important edges.
- Improve defect detection accuracy.

(b) Identify key issues involving noise and edge preservation

The main issues are:

- Noise hides defects.
- Excessive smoothing blurs edges.
- Weak edges may disappear.
- Reduced defect detection accuracy.

(c) Apply an appropriate sequence of techniques

Technique Sequence:

1. **Median Filter** – Remove salt-and-pepper noise while preserving edges.
2. **Canny Edge Detection** – Detect clear and accurate edges.
3. *(Optional)* **Morphological Closing** – Connect small gaps in detected edges.

Result:

- Noise is removed.
- Important edges are preserved.
- Defects become easier to detect.

(d) Justify the order and selection

The selected order is appropriate because:

- **Median Filter** removes noise first without blurring edges.
- **Canny Edge Detection** works better on a clean image and accurately detects boundaries.
- **Morphological Closing** connects any broken edges after detection.

- This sequence improves defect detection accuracy while maintaining image quality.

(e) Clear and structured explanation

Problem:

The inspection system requires both noise removal and edge preservation for accurate defect detection.

Issues:

- Noise affects image quality.
- Edge blurring reduces defect visibility.
- Difficult defect detection.

Solution:

1. Median Filter
2. Canny Edge Detection
3. Morphological Closing (if needed)

Outcome:

- Noise-free image
- Preserved edges
- Clear defect boundaries
- Improved inspection accuracy

Code of Conduct:

*I **DARANI E(1192312120)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

DARANI E

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding ; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand